

Project Description PP1 SoSe2026

Ablating AlphaFold2 Inputs

1 Background

AlphaFold2 (AF2) revolutionized structural biology by accurately predicting 3D structures of proteins from “only” their amino acid sequences¹. In practice, however, the input is not a single sequence at all. For most use cases, AF2 typically relies on a multiple sequence alignment (MSA) of homologous sequences, which provides the network with evolutionary information in the form of co-evolving residue pairs. These co-evolutionary signals are strong structural constraints: they are one of the main reasons why AF2 predicts accurate structures, but they also bias AF2 toward a single “consensus” conformation per sequence.

Many biologically important proteins, however, are not static. Membrane transporters alternate between inward-facing (IF) and outward-facing (OF) states to move substrates across the membrane, and G-protein-coupled receptors (GPCRs) undergo large rearrangements between inactive and active states to initiate signaling cascades². Capturing such conformational heterogeneity is crucial for mechanistic understanding and for structure-based protein design.

Several works have shown that AF2 *can* be coerced into sampling alternative conformations by perturbing its MSA input. Stochastically **subsampling** the MSA to very shallow depths (as few as 16-32 sequences) yields accurate models of both endpoint conformations for a benchmark of transporters and GPCRs². SPEACH_AF³ performs targeted *in silico* **alanine mutagenesis** on interacting residue sets across the MSA to break specific contacts. AFsample2⁴ later generalized this idea by **randomly masking entire columns** of the MSA with the “unknown residue” token X, diluting co-evolutionary signals in an unbiased way.

A complementary suggestion by Ovchinnikov⁵ is to **mask or replace positions** in the **(input) query sequence** itself, rather than the MSA, and observe how the predicted structure responds. Remarkably, even in single-sequence mode without any MSA, more than half of the query residues can be replaced with alanine while the predicted structure remains largely unchanged.

While each of these AF2 ablations has been studied in isolation, their combined effects have yet to be benchmarked in a head-to-head comparison. In this project, we will close this gap by systematically ablating the information that reaches AF2 through both the MSA and the query sequence, and by quantifying how each ablation shifts the conformation ensemble.

¹Jumper, John, et al. "Highly accurate protein structure prediction with AlphaFold." *nature* 596.7873 (2021): 583-589. <https://doi.org/10.1038/s41586-021-03819-2>.

²Del Alamo, Diego, et al. "Sampling alternative conformational states of transporters and receptors with AlphaFold2." *elife* 11 (2022): e75751. <https://doi.org/10.7554/eLife.75751>.

³Stein, Richard A., and Hassane S. Mchaourab. "SPEACH_AF: Sampling protein ensembles and conformational heterogeneity with AlphaFold2." *PLoS computational biology* 18.8 (2022): e1010483. <https://doi.org/10.1371/journal.pcbi.1010483>.

⁴Kalakoti, Yogesh, and Björn Wallner. "AFsample2 predicts multiple conformations and ensembles with AlphaFold2." *Communications biology* 8.1 (2025): 373. <https://doi.org/10.1038/s42003-025-07791-9>.

⁵Ovchinnikov, Sergey. "Ever wondered how many amino acids you can mutate to alanine and AlphaFold2 still predicts same structure?" Post on X/Twitter, 2024. <https://xcancel.com/sokrypton/status/1812769477228200086>.

2 Datasets

We will use the membrane-protein benchmark compiled by del Alamo et al.², consisting of twelve membrane proteins with two experimentally determined structures each: eight whose structures were absent from the AF2 training set, and four additional controls where one conformation is already in the training set. Things to know about this benchmark:

- Targets not present in the AF2 training set (enabling unbiased evaluation):
 - Transporters: *LAT1*, *ZnT8*, *MCT1*, *STP10*, *ASCT2*.
 - GPCRs: *CGRPR*, *PTH1R*, *FZD7*.
- For each target, two experimental structures are available:
 - IF/OF for transporters, e.g., *LAT1* in an IF (PDB: 6IRS, chain B) and an OF (PDB: 7DSQ, chain B) conformation.
 - Inactive/active for GPCRs, e.g., *PTH1R* in an inactive (PDB: 6FJ3, chain R) and an active (PDB: 6NBF, chain R) conformation.
- (Optional) additional targets with *one* conformation already in the AF2 training set, useful as controls where MSA subsampling alone is known to fail:
 - Transporters: *SERT*, *PfMATE*, *MurJ*.
 - GPCRs: *CCR5*.
- The specific AF2 version to be used is the original **monomer** model released with 1 (trained on PDB structures deposited on or before 30 April 2018). Later-trained weights may have one or both of the “not in training set” conformations in their training data and must not be used.
- The dataset is not directly provided as a download. Instead target sequences, truncation boundaries, and reference PDB IDs are described e.g. in the “Materials and methods” section².

3 Exploration Steps

1. Literature

- Familiarize yourself with the role of the MSA in AF2 and how it encodes co-evolutionary constraints.
- Understand how conformational diversity can be induced via MSA subsampling², alanine/column masking of the MSA (SPEACH_AF³, AFsample2⁴), and, potentially, query sequence masking⁵.
- Understand the evaluation protocol: template modeling (TM) score computed with TM-align⁶, principal component analysis (PCA) and root mean square fluctuation (RMSF) of the generated ensemble.

⁶Zhang, Yang, and Jeffrey Skolnick. "TM-align: a protein structure alignment algorithm based on the TM-score." Nucleic acids research 33.7 (2005): 2302-2309. <https://doi.org/10.1093/nar/gki524>.



2. Setup

- Prepare the dataset as outlined in del Alamo et al.².
- The basis of this analysis should be one of the official ColabFold v1.6.1 notebooks:
 - Single-target: `AlphaFold2.ipynb`.
 - Or, alternatively, the batched version: `AlphaFold2_batch.ipynb`.
- Modify ColabFold to support *three* complementary kinds of input perturbations, where “perturb” or “mask” means replacing the affected positions with either the unknown token X or a chosen amino acid (e.g., alanine):
 - Perturb a controllable fraction of positions in the **query sequence**.
 - Perturb a controllable fraction of **columns in the MSA** (excluding first query row).
 - Perturb a controllable fraction of **rows in the MSA** (excluding first query row).
Note: This is probably best achieved through the `custom_msa` option of ColabFold.
- Additionally, follow del Alamo et al.² and implement “**MSA subsampling**” as described in the “Materials and methods” section by setting AF2’s `max_msa_clusters` and `max_extra_msa` parameters before inference.
 - ColabFold implements both parameters via a single `max_msa` argument of the `run` function.
- Replicate the prediction setup of del Alamo et al.² as closely as the Colab budget allows:
 - Disable templates and set the number of recycle steps to one.
 - Skip the final Amber relaxation.
 - Generate 25-50 models per configuration across the 5 available AF2 checkpoints.
- For reference, the official source code of the cited publications is available here:
 - MSA subsampling²: `delalamo/af2_conformations`.
 - SPEACH_AF³: `RSvan/SPEACH_AF`.
 - AFsample2⁴: `wallnerlab/AFsample2`.
 - Notebook with query sequence mutation⁵ code: `replacement_scan.ipynb`.
- For TM-score computation, use the `tmtools` Python package, which wraps TM-align⁶.
- For the PCA analysis, you can follow SPEACH_AF³ and use the ProDy package⁷.

3. Establish baselines

Establish the following baselines on at least two representative targets (i.e., one transporter and one GPCR), preferably on all eight (or twelve) of them. For each baseline and ablation setting, generate an ensemble of 25-50 models and record all predictions along with their predicted Local Distance Difference Test (pLDDT) values. You are free to and probably have to reduce the parameter ranges below. Parameter values that should be prioritized are **bold**.

- **MSA depth**²: Subsample the MSA to depth {16, 32, 64, **128**, 256, 512, 1024, **5120**}.
- **MSA column masking**⁴: Randomly replace (mask) a fraction {5, **15**, 30, 50}% of the MSA columns (excluding the query row) with the X token or e.g., alanine.

⁷Bakan, Ahmet, Lidio M. Meireles, and Ivet Bahar. "ProDy: protein dynamics inferred from theory and experiments." *Bioinformatics* 27.11 (2011): 1575-1577. <https://doi.org/10.1093/bioinformatics/btr168>.



4. Research Question: How redundant is the query sequence for AF2?

After establishing the baselines, we focus on the following ablations to answer how AF2 utilizes the query sequence in presence and absence of MSA information.

- **Query masking⁵:** Mask {5, 15, 50, 90, 100}% of the query sequence while providing full MSA information.
- **Query masking + MSA depth:** Mask {5, 15, 50, 90, 100}% of the query sequence while subsampling the MSA to {16, 32, 64, 128, 256, 512, 1024, 5120} rows.
- **Query masking + MSA column masking:** Mask {5, 15, 50, 90, 100}% of the query sequence while masking up to {5, 15, 30, 50}% of the MSA columns. If possible, ensure that rows masked in the query are not masked in the MSA and vice versa.

5. Research Question: Does AF2 utilize empty rows?

The hypothesis behind this research question is that the latent representation of AF2 benefits from the additional space afforded by informationless “empty rows”. Through the attention mechanism, it should be possible for AF2 to identify rows with comparatively higher information content, which it could potentially “copy” or merge into empty rows for future layers and recycling steps. This would result in more compute per-token when compared to the **MSA depth** setting where the number of rows and size of the latent representation are reduced in lockstep.

- **MSA row masking:** Provide a custom MSA to ColabFold where you mask out all but the first {16, 32, 64, 128, 256, 512, 1024, 5120} rows with the X token or e.g., alanine.

6. Evaluation

For evaluation, follow the procedures described in 2, 3 and 4:

- Scatter plots of TM-score to IF/inactive vs. TM-score to OF/active (Figure. 1, 2; Fig. 4 & 5, 3).
- Scatter plots of PC1 from PCA of C α coordinates across the ensemble against TM-score (Figure. 3, 2). (Optional) Report the variance explained by PC1.
- Scatter plots of PC1 and PC2 (Fig. 4 & 5, 3).
- (Optional) Best-of-ensemble TM-score for each of the two reference states (Fig. 3, 4).
- (Optional) RMSF per residue and its correlation with the experimental C α displacement between the two reference structures.

Then specifically compare:

- **MSA depth vs. MSA column masking vs. Query masking**
- **Query masking vs. Query masking + MSA depth**
- **Query masking vs. Query masking + MSA column masking**
- **MSA depth vs. MSA row masking**
- **MSA column masking vs. MSA row masking**

7. (Optional) Further exploration

- AFsample2⁴ introduces the OC23 and OC > 85 datasets and references a more recent *Transporters*⁸ dataset, for which you could repeat this analysis

⁸Xie, Tengyu, and Jing Huang. "Can protein structure prediction methods capture alternative conformations of membrane transporters?." *Journal of Chemical Information and Modeling* 64.8 (2024): 3524-3536. <https://doi.org/10.1021/acs.jcim.3c01936>.

